Reproducibility Notes

Repository documentation for datasets, scripts, and figure generation

This document explains how the repository supports reproducibility of the thesis analysis. The repository is designed to preserve the input datasets, core scripts, and figure-generation workflow used for the Higgs-portal vector dark matter reinterpretation.

Suggested repository structure:

datasets/ numerical inputs and processed data files
scripts/ Python scripts for conversions and figure generation
figures/ final PNG/PDF figures exported from the scripts
docs/ supporting documentation, including this file

Main software requirements:

- Python 3.x
- NumPy
- Pandas
- Matplotlib
- SciPy, where interpolation or numerical support is needed

Recommended reproduction order:

1. Check datasets/collider_limits.csv and datasets/higgs_constants.csv.
2. Run the BR-to-width conversion script.
3. Run the width-to-coupling script and generate the full coupling plot.
4. Run the threshold-region plotting script.
5. Run the uncertainty-band propagation script for the ATLAS 2022 VBF input.
6. Check direct-detection CSV files and run the sigma_SI overlay plotting script.
7. Compare the generated outputs with the figures retained in the thesis.

Assumptions and limitations:

- The analysis is a phenomenological reinterpretation of published results, not a new detector-level or likelihood-level analysis.
- The collider-derived scattering limit depends on the chosen Higgs-portal vector dark matter framework and adopted numerical constants.
- The ATLAS 2022 VBF uncertainty bands are propagated only for the source where the required expected +/-1 sigma and +/-2 sigma values were available in usable form.
- The PandaX-4T dataset is digitized from a published figure and should be treated as an approximate reconstruction rather than an official numerical table.

Archive statement: The repository is intended to provide a transparent record of the datasets and scripts used in the thesis. It should be cited or linked in the thesis methodology and Appendix I as the complete code and data archive supporting the selected code excerpts shown in the appendices.